

SAT UZBEKISTAN

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Module-1

1.

At a nature preserve, a wildlife biologist counted ducks from an observation deck at the same time each day for 41 days. The table summarizes the resulting data set, data set A.

Number of ducks	Number of days
0	1
1	7
2	8
3	9
4	8
5	7
18	1

The data value 18 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 40 data values. Which statement best compares the median of data set A and the median of data set B?

- ☐ A) There is not enough information to compare the medians of the two data sets.
- ☐ B) The median of data set B is equal to the median of data set A.
- ☐ C) The median of data set B is greater than the median of data set A.
- ☐ D) The median of data set B is less than the median of data set A.

2.

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.6 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

- ☐ A) $3.0x + 1.2y = 3.6$
- ☐ B) $0.33x + 0.83y = 3.6$
- ☐ C) $1.2x + 3.0y = 3.6$
- ☐ D) $0.83x + 0.33y = 3.6$

3.

A certain town has an area of 13,753,344 square yards. What is the area, in square miles, of this town? (1 mile = 1,760 yards)

☐ A) 7,814.4

☐ B) 2.11

☐ C) 3,708.55

☐ D) 4.44

4.

$$f(x) = 4d(18x + 19) + 20$$

Which of the following represents the x -intercept of the graph of $y = f(x) + 6$ in the xy -plane, where d is a constant?

☐ A) $(-\frac{76d-26}{72d}, 0)$

☐ B) $(6 - \frac{76d+20}{72d}, 0)$

☐ C) $(76d + 26, 0)$

☐ D) $(-\frac{39}{72d+6}, 0)$

5.

In triangle ABC , the measure of angle A is 56° and $AC = 35$. In triangle PQR , the measure of angle P is 56° and $PR = 105$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

☐ A) $AB = 50$ and $PQ = 50$.

☐ B) The measures of angle B and angle R are 32° and 92° , respectively.

☐ C) The measures of angle B and angle Q are 56° and 32° , respectively.

☐ D) $AB = 50$ and $QR = 150$.

6.

A quadratic function gives the estimated length of daylight $d(t)$, in hours, in a certain city t months after March 1, where $0 \leq t \leq 7$. According to the function, the estimated length of daylight is 13.44 hours 5 months after March 1 and the maximum estimated length of daylight is 13.89 hours 3.5 months after March 1. Based on this function, what is the estimated length of daylight, in hours, on March 1?

7.

$$kx^2 - 16x = 27x^2 - 8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

Ⓐ

8.

Rectangles S and R are similar. The ratio of each side length of rectangle S to the corresponding side length of rectangle R is 25 to 49. The ratio of the area of rectangle S to the area of rectangle R is 625 to r . What is the value of r ?

ⒺR:

9.

A freight elevator can hold a maximum weight of 4,640 pounds during one trip. A 160-pound person needs to deliver several boxes using the freight elevator. Some of these boxes weigh 22 pounds each and the others weigh 60 pounds each. Which inequality represents the possible combinations of the number of 22-pound boxes, x , and the number of 60-pound boxes, y , the person can deliver during one trip if only the person and the boxes are on the freight elevator?

☐ A) $60x + 22y \leq 4,640$

☐ B) $22x + 60y \geq 4,480$

☐ C) $60x + 22y \geq 4,640$

☒ D) $22x + 60y \leq 4,480$

10.

A quadratic function C models the coefficient of lift of a flying disc, where x is the angle of incidence, in degrees, and $30 \leq x \leq 45$. The function estimates that the maximum value of the coefficient of lift is 1.74, achieved when the angle of incidence is 41 degrees. Which equation could define C ?

☐ A) $C(x) = -0.005(x - 1.74)^2 + 41$

☐ B) $C(x) = 0.005(x + 1.74)^2 + 41$

☒ C) $C(x) = -0.005(x - 41)^2 + 1.74$

☐ D) $C(x) = 0.005(x + 41)^2 + 1.74$

11.

The measure of angle K is $\frac{\pi}{3(17)}$ radians. The measure of angle L is 17 times the measure of angle K . What is the measure, in degrees, of angle L ?

12.

The density of a certain type of marble stone is 2.2000 grams per cubic centimeter. If a sample of this type of stone is in the shape of a sphere with a diameter of 33.000 centimeters, what is the mass of this sample, in grams, to the nearest whole number? (Use 3.14159 for π .)

13.

The length of the diagonal of a square is $\frac{188\sqrt{2}}{2}$ units. What is the area, in square units, of the square?

14.

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A)

Value	Frequency
50	7
60	6
70	6
80	7

B)

Value	Frequency
50	4
60	5
70	6
80	7

C)

Value	Frequency
50	6
60	6
70	6
80	6

D)

Value	Frequency
50	8
60	5
70	5
80	8

15.

At the beginning of an experiment, the temperature of a liquid is 23 degrees Celsius ($^{\circ}\text{C}$). During the first 4.0 minutes of the experiment, the temperature of the liquid increases at an average rate of 7.5°C per minute. Then, the temperature of the liquid increases at a constant rate of 2.6°C per minute. If the temperature of the liquid reaches 61°C x minutes after the beginning of the experiment, where $x > 4.0$, which equation represents this situation?

- ☐ A) $61 = 2.6x + 23$
- ☐ B) $61 = 2.6(x - 4.0) + 30$
- ☐ C) $61 = 2.6x + 53$
- ☒ D) $61 = 2.6(x - 4.0) + 53$

16.

The function f is defined by $f(x) = 15x^5$. The graph of $y = f(-x) + c$ in the xy -plane, where c is a positive integer constant, has an x -intercept at $(r, 0)$ and a y -intercept at $(0, t)$, where r and t are constants. Which of the following must be true about r and t ? ER:

- ☐ A) $r < 0$ and $t > 0$
- ☐ B) $r < 0$ and $t < 0$
- ☐ C) $r > 0$ and $t < 0$
- ☒ D) $r > 0$ and $t > 0$

17.

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$900 and did not make any other deposits or withdrawals. 4 months after Bennett opened the account, the balance had increased by 0.6% of the original balance. 6 months after Bennett opened the account, the balance had increased by an additional 0.3% of the balance at the end of the first 4 months. Every 2 months after the first 6 months, the balance had increased by an additional 0.3% of the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

- ☒ A) $B(x) = 905.40(1.003)^{\frac{x}{2} - 4}$
- ☐ B) $B(x) = 905.40(1.003)^{2x - 8}$
- ☐ C) $B(x) = 905.40(1.003)^{2x - 4}$
- ☐ D) $B(x) = 905.40(1.003)^{\frac{x}{2} - 4}$

18.

$$\sqrt{x^2 - 15x + 175} = x\sqrt{11}$$

What are all possible solutions to the given equation?

☐ A) $\frac{7}{2}$ and -5

☒ B) $\frac{7}{2}$

☐ C) $-\frac{7}{2}$ and 5

☐ D) 5

19.

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 63 units, the length of $\overline{LK}$ is 21 units, and the area of triangle RST is 819 square units, what is the length of $\overline{KT}$, in units?

20.

If $\frac{x+6}{5} = \frac{x+6}{11}$, the value of $x + 6$ is between which of the following pairs of values?

☐ A) -7 and -5

☐ B) 8 and 11

☐ C) 4 and 7

☒ D) -4 and 4

21.

The population of a certain city doubled every 75 years from 1658 to 1958. The population of this city was 160,000 in 1958. What was the population of this city in 1658?

22.

For groups of 25 or more people, a museum charges \$21 per person for the first 25 people and \$14 for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

A. $f(n) = 14n + 175$

B. $f(n) = 14n + 525$

C. $f(n) = 35n - 350$

D. $f(n) = 14n + 21$

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Module-2

1.

The cost of renting a backhoe for up to **10** days is **\$270** for the first day and **\$135** for each additional day. Which of the following equations gives the cost y , in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$

2.

A driving school charges \$15 for the first lesson a student takes in a month and \$9 for each additional lesson that month. Which equation describes this situation, where y is the total cost (in dollars) of x lessons for one month, and $x > 0$?

A) $y = x + 15$

B) $y = 9x + 15$

C) $y = 9(x - 1) + 15$

D) $y = (x - 1) + 15$

3.

For groups of 30 or more hikers, a guide service charges \$25 per person for the first 30 hikers and \$16 per person for each additional hiker beyond 30. Which function f gives the total charge (in dollars) for a group of n hikers, where $n \geq 30$?

A) $f(n) = 16n + 270$

B) $f(n) = 25n + 16$

C) $f(n) = 16n + 350$

D) $f(n) = 16n + 750$

4.

The cost of renting a portable generator is \$80 for the first day and \$30 for each additional day. Which of the following functions $G(d)$ gives the total cost (in dollars) of renting the generator for d days, where d is a positive integer?

A) $G(d) = 30d + 30$

B) $G(d) = 30d + 80$

C) $G(d) = 80d - 30$

D) $G(d) = 50 + 30d$

5.

The cost to rent a film studio from Company X is \$1000 for the first 3 hours, plus an additional \$60 per hour for each hour after the first 3. If the total cost to rent the studio for t hours (where $t > 3$) is \$1300, which equation represents this situation?

A) $1000(t - 3) + 60t = 1300$

B) $1000(2t) + 60t = 1300$

C) $1000 + 60(t - 3) = 1300$

D) $1000 + 60(2t) = 1300$

6.

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

7.

A bird-watching group initially observed 200 birds in a certain area and set a goal of reaching 800 birds. According to their model, the population starts at 200 and increases by 30 birds per week for the first two weeks after the observation. After those two weeks, the population increases by 50 birds per week until the goal of 800 birds is reached. At the end of week w (where $w > 2$) after the initial observation, let $p(w)$ be the predicted number of birds still needed to reach the 800-bird goal. Which of the following functions correctly represents $p(w)$?

A) $p(w) = 800 - 50w$

B) $p(w) = 750 + 50w$

C) $p(w) = 640 - 50w$

D) $p(w) = 260 + 50w$

8.

A club sold tickets to a carnival to raise funds. Each child ticket sold for \$8, and each adult ticket sold for \$12. The club earned a total of \$576 from selling these tickets. The equation

$$8x + 12y = 576$$

represents this situation. Which of the following is the best interpretation of y in this context?

A) The number of adult tickets sold

B) The number of child tickets sold

C) The price, in dollars, of each adult ticket

D) The price, in dollars, of each child ticket

9.

Hassan put up wire fencing along each edge of a rectangular garden. The garden has a length of x feet and a width of y feet. He used a total of 36 feet of wire fencing.

Which equation represents this situation?

A. $2x + 2y = 36$

B. $x + y = 36$

C. $2xy = 36$

D. $xy = 36$

10.

A professor is creating an exam worth 32 points in total. The exam will have 1-point questions and 3-point questions. Let x be the number of 1-point questions, and y the number of 3-point questions.

Which equation represents this situation?

A) $3xy = 32$

B) $3(x + y) = 32$

C) $x + 2y = 32$

D) $x + 3y = 32$

11.

A bakery calculates its monthly profit by subtracting its monthly operating costs from its monthly sales revenue. For a month in which x loaves of bread are produced and sold, the equation

$$12,000 = 3.75x - 9,200$$

represents the situation. Which statement is the best interpretation of $3.75x$ in this context?

- A) The total monthly sales revenue, in dollars, from selling x loaves
- B) The monthly sales revenue, in dollars, from each loaf sold
- C) The total monthly cost, in dollars, of baking x loaves
- D) The monthly cost, in dollars, for each loaf baked

12.

A group loads money into an account to play arcade games. Each time they play a game, the same amount of money is deducted from the account. The function

$$f(m) = 25 - 4m$$

gives the amount of money (in dollars) left in the account after m games have been played. Which of the following represents the amount of money, in dollars, deducted each time the group plays a game?

- A) $4m$
- B) 4
- C) 25
- D) $25 - 4m$

13.

A group of horticulture students studied the relationship between the length of a plant's largest leaf and the plant's total height. They derived the equation

$$H = 2.32L + 33.5$$

which describes the relationship between the leaf length L , in centimeters, and the plant's estimated height H , in centimeters. Which of the following is the best interpretation of the number 2.32 in this context?

- A) The increase in the plant's estimated height, in centimeters, for each 1-cm increase in leaf length
- B) The increase in the plant's leaf length, in centimeters, for each 1-cm increase in the plant's estimated height
- C) The plant's estimated height, in centimeters, when $L = 0$
- D) The difference, in centimeters, between leaf length and plant height

14.

A partially filled container holds 15 milliliters of water and is placed under a faucet that drips 0.05 milliliters of water every 5 seconds. Until the container is full, which of the following can be used to represent the volume v , in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 5?

- A) $v = 0.01t + 15$
- B) $v = 0.02t + 15$
- C) $v = 0.05t + 15$
- D) $v = 5t$

15.

A factory machine makes three kinds of parts. On a particular day, it makes 8-inch parts, 5-inch parts, and 3-inch parts. The number of 8-inch parts produced is 2 times the number of 5-inch parts, represented by n , and the machine makes 10 of the 3-inch parts. Altogether, the machine produces 70 parts that day. Which equation best represents this situation?

A) $8 \cdot (2n) + 5n + 3 \cdot 10 = 70$

B) $8n + 5n + 3n = 70$

C) $2n + 10 = 70$

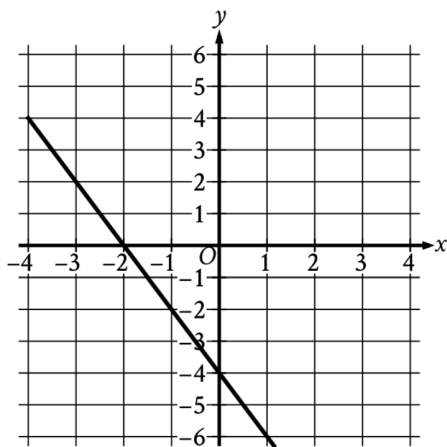
D) $3n + 10 = 70$

16.

The function $G(x) = \frac{9}{5}(x - 273) + 32$ converts a temperature of x kelvins into degrees Fahrenheit. If a temperature increases by 3.50 kelvins, by how many degrees Fahrenheit does the temperature increase?

17.

The graph of $y = f(x) - 11$ is shown.



Which equation defines the linear function f ?

A. $f(x) = -13x - 11$

B. $f(x) = -2x + 7$

C. $f(x) = -13x + 7$

D. $f(x) = -2x - 11$

18.

The function p is defined by

$$p(x) = a((x + 5)^2 - b)((x + 5)^2 - c)$$

where a , b , and c are constants. In the xy -plane, the graph of $y = p(x)$ passes through the points $(-3, 45)$ and $(3, 186)$. What is the value of $p(-7) + p(-13)$?

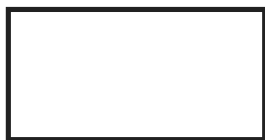
19.

The function q is defined by

$$q(x) = a|x - 9|^2 - 53|x - 9| + b$$

where a and b are constants, and $a > b > 53$. If $q(539) = h$ and $q(-521) = k$, where h and k are constants, what is the value of

$$32(-539)^{h-k} + 347(9)^{k-h}?$$



20.

$$\begin{aligned} 7x + 6y &= 5 \\ 28x + 24y &= 20 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$
- B. $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$
- C. $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$
- D. $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$

21.

One of the two equations in a system of linear equations is given:

$$3x = 15y + 9$$

Which equation could be the second equation in this system, if the system has no solution?

- A) $x = 5y + 3$
- B) $x = 5y + 1$
- C) $x = 2y + 3$
- D) $x = 10y + 3$

22.

$$25x = 3750y + 1000$$

One of the two equations in a system of linear equations is given. The system has no solution.

Which equation could be the second equation in this system?

- A) $\frac{1}{25}x = 5y$
- B) $\frac{1}{25}x = 150y + 60$
- C) $x = 5y$
- D) $x = 150y + 60$